DAY 9 | PRE-READ

Borrowing Tools Safely

A light read before Day 9. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

Yesterday you wrote JavaScript and learned how it handles waiting. Today you build a real web server in JavaScript with Node.js and a popular tool called Express, then test it without writing any screens.

1. JavaScript leaves the browser

JavaScript was born inside web browsers. **Node.js** lets the same language run on servers and laptops. That means one language for the screens and the server, which is why many teams use it.

2. Borrowing tools: npm

Nobody builds everything from scratch. **npm** is a huge free library of ready-made JavaScript tools, like an app store for code. A file called **package.json** is your shopping list: it says which tools your project needs.

3. Checking what you borrow NEW (v4)

When you borrow a ladder from a neighbour, you check it is safe before you climb. Code packages are the same. One package often brings many others with it. A file called the **lockfile** records exactly which versions you used, so everyone gets the same thing. A tool called **npm audit** checks your packages against a list of known security problems.

Key idea: every package you install is code you trust. Install only what you need, record exact versions, and check them regularly.

4. A web server in a few lines

Express is a small tool for building web servers in Node.js. You will write the same kind of API you built with FastAPI: create, read, update and delete items, with clear status codes.

5. Testing without building a screen

How do you test an API before anyone builds the app screens? You use **Postman**, a free tool where you type an address, pick GET or POST, press Send, and see the reply.

Before class: install Postman (desktop or web) and, for VS Code users, Node.js LTS from nodejs.org.

6. Think about these before class

- Why might two laptops get different results from the same project if versions are not recorded?
- Would you install a package with a name that is one letter different from a popular one? Why not?
- Why is it useful to test an API before the app screens exist?

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